UNIVERSITY STUDENT ANALYTICS

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PART 1 — Data Exploration & Understanding

Dataset	University Student Dataset — Post-Cleaning
Records	3,000 students · 40 columns · 3 Report tabs
Scope	Academic Performance · Dropout Risk · Financial · Career · Engagement
Prepared by	Salma Rahhal

3,000 Students	40 Columns	11 Columns with Missing Data	3 Report Tabs

Overview about the data:

This dataset represents university student lifecycle analytics, combining academic performance, student engagement, financial background, wellbeing, and future employability indicators.

Grain: One row = One student

The dataset contains recent, up-to-date operational data where all students are currently enrolled and still studying.

PART 2 — Data Cleaning & Preparation using Python

Before building any dashboard or analysis, the dataset was profiled and cleaned. Below are all steps taken, with exact details on what was found and how it was resolved.

Please find the attached Jupyter Notebook which highlight the code done with Python

Step 0 — Initial Dataset Profiling

Property	Finding
Total Rows	3,000 student records
Total Columns	40 columns — 22 numeric, 18 categorical

Duplicate Rows	0 duplicates found — dataset is clean in this dimension
Columns with Missing Data	11 columns — ranging from 5.2% to 15.9% missing
Data Type Issues	None — all columns loaded with correct data types (int64, float64, object)
Outlier Columns	3 columns with IQR-detected outliers: Study_Hours_Per_Week, Weekly_Work_Hours, Family_Income

Step 1 — Missing Value Analysis & Treatment

Eleven columns contained missing values. Each was handled with a strategy appropriate to its data type and business meaning.

Column	Missing	%	Treatment Strategy	Reason
Starting_Salary_Offer	478	**15.9%**	Fill 0 if Employment_Offer='No'. Fill median of 'Yes' group ($72,898) if Yes.	*Conditional logic — salary only exists when employed*
SAT_Score	343	**11.4%**	Median imputation: 1,245	*Numeric, right-skewed — median preferred over mean*
Scholarship_Amount	310	**10.3%**	Fill 0 if Scholarship_Status='No'. Fill median of 'Yes' group ($25,554) if Yes.	*Conditional — no scholarship means amount is 0*
Weekly_Work_Hours	279	9.3%	Median imputation: 15 hrs	*Numeric continuous — median imputation*
Family_Income	262	8.7%	Median imputation: $162,478	*Right-skewed distribution — median preferred*
High_School_GPA	225	7.5%	Median imputation: 3.24	*Bounded 0–4 scale — median safe*
Distance_From_Campus_km	194	6.5%	Median imputation: 24.5 km	*Continuous spatial variable*
Attendance_Rate	191	6.4%	Median imputation: 80.6%	*Bounded 0–100% — median preferred*
Study_Hours_Per_Week	191	6.4%	Median imputation: 27 hrs	*Continuous behavioral variable*
English_Proficiency_Score	166	5.5%	Median imputation: 90	*Numeric score — median preferred*

Parental_Education_Level	157	5.2 %	Mode imputation: 'PhD' (most frequent category)	*Categorical — mode is correct approach*

After treatment: 0 missing values remain across all 3,000 records and 40 columns. Total cells imputed: 2,796.

Step 2 — Duplicate Detection

✓ **No Duplicate Rows Found**
All 3,000 rows are unique records. Student_ID values are unique sequential identifiers (1–3,000) confirming each row represents a distinct student. No action required.

Step 3 — Data Type Validation

All 40 columns loaded with the correct data types. No type conversions were necessary. Validated types are:

Data Type	Columns
int64 — Integer	Student_ID, Age, Year_of_Study, Credits_Completed, Credits_This_Semester, Internship_Count, Clubs_Joined, Online_Courses_Taken, Publications_Count, Graduation_Expected_Year
float64 — Decimal	GPA, High_School_GPA, SAT_Score, Scholarship_Amount, Distance_From_Campus_km, Attendance_Rate, Study_Hours_Per_Week, Weekly_Work_Hours, English_Proficiency_Score, Family_Income, Dropout_Risk_Score, Starting_Salary_Offer
object — Categorical	Gender, Nationality, Program, Major_Specialization, Enrollment_Type, Scholarship_Status, Accommodation_Type, Part_Time_Job, Internship_Completed, Leadership_Role, Research_Participation, Exchange_Program, Financial_Aid, Parental_Education_Level, Health_Insurance, Mental_Health_Support_Used, Academic_Probation, Employment_Offer

Step 4 — Outlier Detection & Treatment

The IQR (Interquartile Range) method was used to identify outliers in all numeric columns. Three columns contained outliers beyond the 1.5×IQR fence:

Column	Outliers Found	Valid Range	IQR Bounds	Treatment Applied
Study_Hours_Per_Week	4 values	[5, 89]	[0, 71]	Winsorized — capped at 71 hrs
Weekly_Work_Hours	7 values	[0, 75]	[0, 44.5]	Winsorized — capped at 44.5 hrs
Family_Income	17 values	[$20K, $800K]	[0, $447K]	Winsorized — capped at $447,504

Winsorization (capping) was chosen over deletion because these outliers represent real students — extreme but plausible values. Deleting them would lose valid records. Capping retains the records while reducing distortion in averages and models.

Step 5 — Final Validation

Metric	Before Cleaning	After Cleaning	Status
Total Rows	3,000	**3,000**	✅ No rows lost
Total Columns	40	**40**	✅ Enriched
Missing Values	2,796 cells	**0**	✅ Fully imputed
Duplicates	0	0	✅ Clean
Outliers	28 values in 3 columns	**28 winsorized**	✅ Capped, not deleted

PART 3— Data Quality Check

Next, a **data quality assessment** was conducted by validating all dataset dimensions column by column, The following DQ dimensions were evaluated:

1. **Completeness :** Checked for missing, null, or blank values across all fields.
2. **Accuracy :** Verified that values fall within expected business and academic ranges (e.g., GPA limits, attendance percentages).
3. **Consistency:** Ensured uniform formatting, standardized categorical values, and alignment across related attributes.
4. **Validity:** Confirmed data types, formats, and allowed value domains comply with predefined rules.
5. **Uniqueness:** Verified absence of duplicate student records and confirmed unique student identifiers.
6. **Integrity:** Validated logical relationships between columns (e.g., credit completion aligned with academic level).
7. **Timeliness:** Confirmed the dataset reflects current operational data where all students are actively enrolled.

Result: All columns successfully passed the data quality validation, confirming the dataset is complete, consistent, and ready for analysis.

PART 4 — Interactive Dashboard Design using Tableau

The dashboard is organized into 3 main tabs, each answering a distinct set of business questions. Every tab has its own KPI cards, primary charts, and supporting analytics. Below is the complete specification for each tab.

TAB 1 — Executive Overview

High-level institution snapshot — designed for leadership and academic leadership meetings

Purpose: Give any viewer an instant understanding of the institution's health across all dimensions in one view. This is the homepage of the dashboard — every metric here links to a deeper tab.

TAB 2 — Academic Performance

GPA, Attendance, SAT, Credits, Study Hours analysis by program and cohort

TAB 3 — Career Outcomes

Employment offers, starting salaries, and graduation pipeline

PART 5 — Key Findings from the Analysis

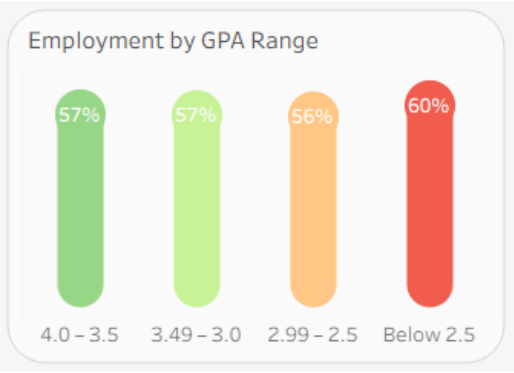
These are the most important, actionable, and sometimes counterintuitive insights discovered through the data analysis. They are organized by theme, each with the supporting data and a recommended action.

📚 Key Findings

F1

GPA doesn't get jobs. Students with GPA < 2.5 have the highest employment rate (59.4%) — more than honor students (56.8%).

The employment rate across GPA bands is surprisingly flat: even the lowest GPA band (< 2.0) achieves a 53.3% employment offer rate,GPA Barely Predicts Employment, Students with GPA < 2.0 Have 53.3% Employment Rate vs 56.8% for Top GPA,



F2

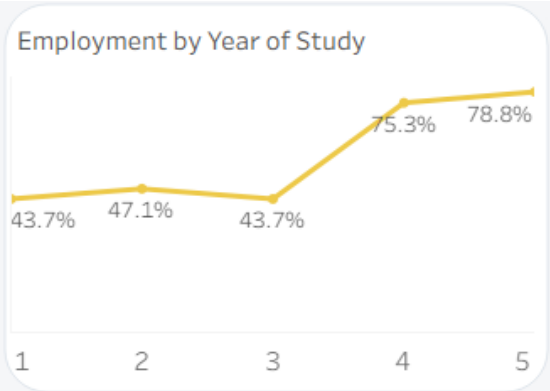
SAT Score Has Near-Zero Correlation With University GPA (r = −0.001)

High school test scores (SAT) do not predict how students will perform at university. Students who scored 800–1,000 on the SAT perform statistically identically to students who scored 1,450+. This suggests the institution's admissions process should not over-weight SAT scores in predicting future academic success.

-0.001
Correlation SAT vs GPA

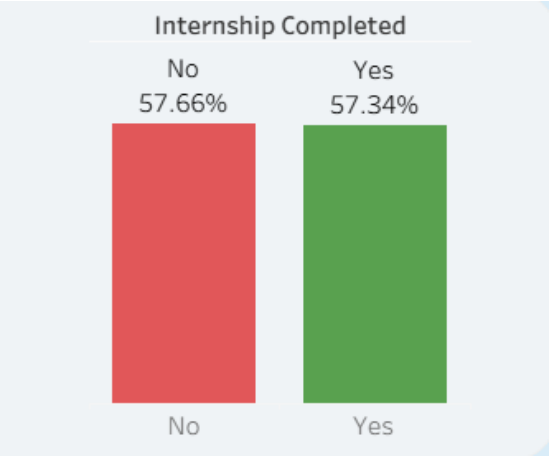
F3

Year 4 & 5 are peak hiring years. Employment jumps from 43.7% in Year 1 to 78.8% in Year 5 — a 35pp leap.



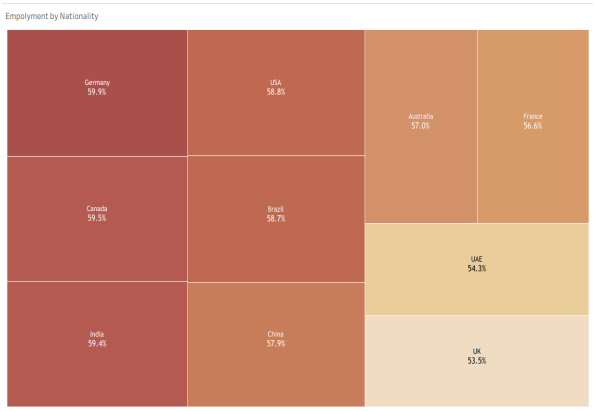
F4

Completing an Internship Provides No Employment Advantage (57.3% vs 57.7%)
Students who completed internships have a 57.3% employment rate — actually 0.4pp lower than students who did not complete an internship (57.7%). Similarly, research participation and leadership roles each show less than 0.5pp difference in employment rates. Traditional 'resume builders' have no measurable impact on employment outcomes in this dataset.



F5

Germany and Canada Produce the Highest Employment Rates by Nationality (59.9% and 59.5%)
German students achieve the highest employment rate (59.9%) and UK students the lowest (53.5%) — a 6.4pp gap. UAE students rank second-to-last (54.3%). This may reflect differences in job-seeking networks, visa status, language skills, or career preparation across nationality groups.



F6

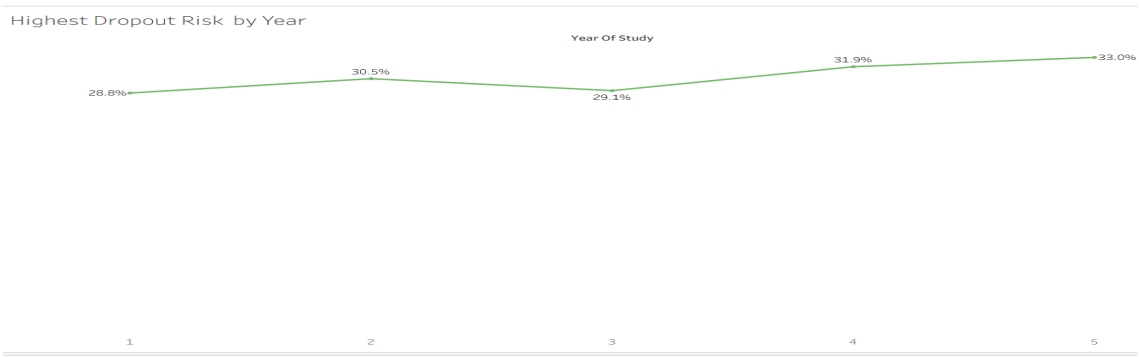
AI Is the Riskiest Program Despite Having the Highest Average GPA (3.04)

AI students have the highest dropout risk (35.2%, +4.5pp above baseline) yet also the highest average GPA. This paradox means AI students' risk is not driven by academic underperformance. Non-academic factors — curriculum intensity, career uncertainty, workload complexity — are the most likely drivers. Standard academic support will not help this cohort.

F7

Year 5 Students Have the Highest Dropout Risk (33.0%) — Final-Year Pressure Is Real

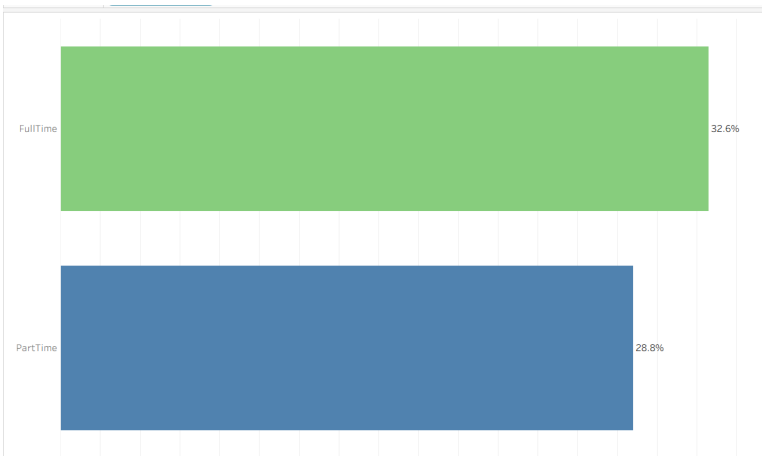
Dropout risk increases progressively from Year 1 (28.8%) to Year 5 (33.0%). Year 4 is an inflection point (31.9%) and Year 5 is the peak. The assumption that students who survived to final year are stable is false. Final-year thesis, dissertation, and career transition pressures create measurable risk escalation.



F8

Full-Time Students Are Higher Risk Than Part-Time (32.6% vs 28.8%)

Counterintuitively, full-time enrolled students show significantly higher dropout risk than part-time students. This may reflect the higher cognitive and time demands of full-time study. Part-time students may have more manageable workloads despite also working jobs.



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